

LMS Rebuild & Enhancement Project Documentation

This document provides a comprehensive overview of the design system, directory structure, implemented features, and modifications applied to the Learning Management System (LMS) web application.

1. Project Overview & Architecture

The LMS frontend application is built using a modern React single-page application (SPA) architecture, optimized for responsive layouts and clean component reuse.

- **Frontend Core Library:** React (v18+)
- **Build Tool / Bundler:** Vite (supporting fast hot module reloading and optimized Rollup packaging)
- **Styling Framework:** Tailwind CSS (for rapid utility-first UI design)
- **Icon Library:** Lucide React (for standard interface icons)
- **Routing:** React Router DOM (v6+ for client-side navigation)
- **State Management:** React Context API (custom `AuthContext` for user roles and sessions)

2. Directory & File Structure

Below is the verified structural schema of the React application source code:

```
Frontend/
├── public/                # Static assets (logos, favicon)
├── src/
│   ├── assets/           # Global visual resources (thumbnails, images)
│   ├── components/       # Reusable UI widgets and layout headers
│   │   ├── common/
│   │   │   ├── Breadcrumb.jsx
│   │   │   ├── Footer.jsx
│   │   │   ├── Navbar.jsx
│   │   │   └── RoleBasedSidebar.jsx
│   │   └── ui/
│   │       ├── Badge.jsx
│   │       ├── Button.jsx
│   │       ├── Card.jsx
│   │       ├── Input.jsx
│   │       ├── ProgressBar.jsx
│   │       └── ProtectedRoute.jsx
│   ├── context/
│   │   └── AuthContext.jsx # Global auth, session, and role state provider
│   ├── layouts/         # Template wrappers for role-based viewports
│   │   ├── AdminLayout.jsx
│   │   ├── InstructorLayout.jsx
│   │   ├── MainLayout.jsx
│   │   └── StudentLayout.jsx
│   ├── pages/           # Portal viewports split by user authorization level
│   │   ├── admin/
│   │   │   ├── AdminDashboard.jsx
│   │   │   ├── AdminPayment.jsx
│   │   │   ├── AdminProfile.jsx
│   │   │   ├── CourseManagement.jsx
│   │   │   ├── Earnings.jsx
│   │   │   ├── Reports.jsx
│   │   │   ├── Settings.jsx
│   │   │   └── UserManagement.jsx
│   │   ├── instructor/
│   │   │   ├── CreateCourse.jsx
│   │   │   ├── Earnings.jsx
│   │   │   ├── InstructorDashboard.jsx
│   │   │   ├── InstructorProfile.jsx
│   │   │   └── ManageCourses.jsx
│   │   ├── public/
│   │   │   ├── CourseDetails.jsx
│   │   │   ├── CourseListing.jsx
│   │   │   ├── Landing.jsx
│   │   │   ├── Login.jsx
│   │   │   └── Register.jsx
│   │   └── student/
│   │       ├── Certificates.jsx
│   │       ├── EditProfile.jsx
│   │       ├── LearnCompleted.jsx
│   │       ├── LearnLesson.jsx
│   │       ├── Payments.jsx
│   │       ├── StudentDashboard.jsx
│   │       └── StudentProfile.jsx
│   ├── utils/
│   │   └── mockData.js    # Data store representing students, courses, and stats
│   ├── App.jsx           # Main routing registry
│   ├── index.css         # Global CSS variables and tailwind layer definitions
│   └── main.jsx          # React root DOM element mounter
├── package.json          # Dependency manifest
├── tailwind.config.js    # Styling scale adjustments
└── vite.config.js        # Vite build options configuration
```

3. Implemented Features & Rebuild Work

🏠 Student Portal Layout

- **Student Dashboard:** Includes visual summary cards for ongoing/completed courses, a learning hours display, next deadline listings, and monthly progress charts.
- **Dynamic Course Learning:** The lesson page supports inline lectures, materials download blocks, course progression metrics, and an active inline quiz panel.
- **Explore Courses:** A custom course catalog view is registered under `/student/explore`, letting enrolled students browse and select new academic courses directly without losing the side navigation panel.
- **Certificates:** Allows students to view, filter, and download successfully earned course diplomas.
- **Tuition Payments:** Enables student invoices auditing, transaction histories tracking, and simulation checkout processed via mock handlers.

👤 Instructor Portal Layout

- **Instructor Dashboard:** Summarizes active learners count, total earnings, active course listings, and lists recent student enrollments.
- **Course Wizard & Management:** Instructors can build courses via multi-step wizards and update lessons or drafts in the custom `ManageCourses` screen.
- **Reports & Earnings Tracking:** Tracks student academic performance metrics and payout history.

🔑 Admin Portal Layout

- **Admin Dashboard:** Real-time server performance statistics and system telemetry overview.
- **Revenues Audit:** The payment screen (`/admin/payments`) provides transparency into gross platforms transactions, processing fees, and net payouts.
- **Users Administration:** Oversees user accounts registry, role groups, and active statuses.

4. Applied UX Enhancements & Fixes

1. Public/Dashboard Nav bars Swap:

- Resolved the issue where a logged-in student would see dashboard greetings on public pages. The system now dynamically renders the public navbar layout when accessing `/` or `/courses`, displaying a "Go to Dashboard" button instead of "Login/Register" for active sessions.

2. Student Navigation Bar Upgrades:

- Integrated a Search bar, Settings icon, and Profile Avatar into the right-hand corner of the student dashboard header, ensuring seamless access to account adjustments.

3. Card Text Contrast Correction:

- Fixed text element overrides where account balances (e.g., LKR 148,500.00) on gradient cards rendered in dark blue. Changed tags from `h2` to `div` to bypass global stylesheet overrides, restoring clean white readability.

4. Guest Redirect Workflow:

- Bypassed empty protected route redirects for unauthenticated sessions. Guests clicking "Buy Now" on public pages are now smoothly redirected to the `/login` portal.

5. Visual Cleanup:

- Replaced raw text indicators like "R" on instructor course cards with professional Lucide `Trash2` icons. Removed all empty template files to clean the codebase structure.